

# *Finance Course*

*Session 6 – 09/03/2026*



## *Value*



**ALBERTSCHOOL**

# Introduction

## **Semester 1: Introduction to Financial Markets**

- 1.1 Presentation of the financial markets (4h)
- 1.2 Mechanisms of financial markets (2h)
- 1.3 Risk in finance (2h)
- 1.4 Equities (2h)
- 1.5 Bonds (2h)
- 1.6 Options (2h)
- 1.7 Role of data in the financial markets (2h)
- 1.7 Financial market regulation (2h)
- 1.8 Case study (4h)

## **Semester 2 : Value**

- 2.1 Reminder of accounting principles (2h)
- 2.2 Financial Analysis (2h)
- 2.3 Measure of creation value (2h)
- 2.4 Cost of capital (2h)
- 2.5 Financial structure (2h)
- 2.6 Methods of valuing a business (8h)
- 2.7 Case Study (4h)
- 2.8 Intervention (2h)

**Comprendre la valeur d'une entreprise au regard de son risque inhérent**

# Methods of valuing a business

## Part 2

# Methods of valuing a business

| Equity Value                             | Indirect method                                                | Direct Method                           |
|------------------------------------------|----------------------------------------------------------------|-----------------------------------------|
| Intrinsic method (discounted cash flows) | Discounting cash flows – net debt                              | Discounting FCFE (or dividends)         |
| Comparative method (multiples)           | Operating income multiple $\times$ operating income – net debt | Net income multiple $\times$ net income |

# Valuation multiples

## Indirect Approach

This consists of calculating the multiples of comparable companies (listed or transactional) starting from the equity value and adding net financial debt in order to obtain the enterprise value (value of the economic asset).



## Direct Approach

This consists of calculating the multiples of comparable companies (listed or transactional) based only on the equity value.



# Valuing a business with multiples

# The Multiples methods

The multiples method is based on three main principles:

- The company is valued as a whole
- The company is valued as a multiple of its earnings capacity
- Markets are assumed to be in equilibrium, therefore comparisons are justified

This approach does not focus on the value of assets and liabilities but rather on the profitability resulting from their use. The level of multiples depends on the company's profitability levels and growth prospects.

The higher the profitability levels and growth prospects, and the lower the risk of the market in which the company operates, the higher the expected multiples will be.

This approach is comparative, meaning it is a relative valuation method, evolving according to market conditions and the valuation levels of comparable companies.

The multiples method relies on a sample of listed companies or companies that have recently been sold. We therefore refer to trading multiples or transaction multiples.

# Setting up a list of comparables

To perform a valuation using the comparables method it is necessary to determine a relevant sample of companies comparable to the company we wish to value.

Criteria for a relevant sample:

1. Sector
2. Activity
3. Size
4. Geographic presence
5. Profitability
6. Growth



# Example

Determine a sample for the company TOTAL ENERGIES using the following companies:

| Entreprise        | Capitalisation boursière (En Mds €) | Pays d'origine       | Activité                                         |
|-------------------|-------------------------------------|----------------------|--------------------------------------------------|
| Saudi Aramco      | 2000                                | Arabie Saoudite      | Exploration, production, raffinage, distribution |
| ExxonMobil        | 415                                 | États-Unis           | Exploration, production, raffinage, distribution |
| Royal Dutch Shell | 205                                 | Pays-Bas/Royaume-Uni | Exploration, production, raffinage, distribution |
| Chevron           | 283                                 | États-Unis           | Exploration, production, raffinage, distribution |
| BP                | 100                                 | Royaume-Uni          | Exploration, production, raffinage, distribution |
| TotalEnergies     | 140                                 | France               | Exploration, production, raffinage, distribution |
| Gazprom           | 41                                  | Russie               | Exploration, production, distribution            |
| PetroChina        | 221                                 | Chine                | Exploration, production, distribution            |
| ConocoPhillips    | 132                                 | États-Unis           | Exploration, production, raffinage, distribution |

# Example

Determine a sample for the company TOTAL ENERGIES using the following companies:

| Entreprise        | Capitalisation boursière (En Mds €) | Pays d'origine       | Activité                                         | Commentaires               |
|-------------------|-------------------------------------|----------------------|--------------------------------------------------|----------------------------|
| Saudi Aramco      | 2000                                | Arabie Saoudite      | Exploration, production, raffinage, distribution | Trop grand                 |
| ExxonMobil        | 415                                 | États-Unis           | Exploration, production, raffinage, distribution | Ok                         |
| Royal Dutch Shell | 205                                 | Pays-Bas/Royaume-Uni | Exploration, production, raffinage, distribution | Ok                         |
| Chevron           | 283                                 | États-Unis           | Exploration, production, raffinage, distribution | Ok                         |
| BP                | 100                                 | Royaume-Uni          | Exploration, production, raffinage, distribution | Ok                         |
| TotalEnergies     | 140                                 | France               | Exploration, production, raffinage, distribution |                            |
| Gazprom           | 41                                  | Russie               | Exploration, production, distribution            | Russie / pas même activité |
| PetroChina        | 221                                 | Chine                | Exploration, production, distribution            | Chine / pas même activité  |
| ConocoPhillips    | 132                                 | États-Unis           | Exploration, production, raffinage, distribution | Ok                         |

# Valuation multiples

Two main categories of multiples can be distinguished:

1/ Multiples used to determine the enterprise value

- These multiples are based on financial aggregates before the impact of financial expenses
- Example: Revenue / EBE (EBITDA) / REX (EBIT)

This corresponds to the indirect approach.

Example: if we obtain a multiple of 8x EBITDA, this means that the enterprise value equals 8 years of EBITDA (assuming it remains constant).

2/ Multiples used to determine the equity value

- These multiples are based on financial aggregates after the impact of financial expenses
- Example: Net income / P/E ratio (PER)

This corresponds to the direct approach.

The indirect approach is the one most commonly used by analysts.

# Enterprise Value multiples

## **Operating income multiple (EV/EBIT)**

The operating income multiple (EV/EBIT) compares the enterprise value to operating income. It reflects the operating earnings capacity of companies based on recurring normalized operating profitability.

## **EBITDA multiple (EV/EBITDA)**

The EV/EBITDA multiple follows the same logic as EV/EBIT, but removes the impact of depreciation policies, which can vary significantly from one company to another. This ratio is often more relevant for companies with significant depreciation levels (for example, industrial companies).

## **Revenue multiple (EV/Revenue)**

The EV/Revenue multiple ignores the company's profitability and considers only the size of its activity. This multiple can be useful for companies that have not yet reached profitability, or companies with a similar cost structure.

# Example

Based on the following information, determine:

1/The enterprise value (economic asset value)

2/The valuation multiples

- EV / Revenue
- EV / EBITDA
- EV / EBIT

|                                 | Entreprise A | Entreprise B | Entreprise C | Entreprise D |
|---------------------------------|--------------|--------------|--------------|--------------|
| <b>Capitalisation boursière</b> | 16 000       | 3 000        | 4 000        | 36 000       |
| <b>Dette financière nette</b>   | 3 000        | (250)        | 2 000        | 13 000       |
| <b>Chiffre d'affaires</b>       | 16 000       | 1 200        | 5 000        | 40 000       |
| <b>EBE (EBITDA)</b>             | 1 280        | 120          | 400          | 2 400        |
| <b>REX (EBIT)</b>               | 960          | 96           | 300          | 1 600        |

# Example

Based on the following information, determine:

1/ The enterprise value (economic asset value)

|                                 | Entreprise A | Entreprise B | Entreprise C | Entreprise D |
|---------------------------------|--------------|--------------|--------------|--------------|
| <b>Capitalisation boursière</b> | 16 000       | 3 000        | 4 000        | 36 000       |
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| <b>EBE (EBITDA)</b>             | 1 280        | 120          | 400          | 2 400        |
| <b>REX (EBIT)</b>               | 960          | 96           | 300          | 1 600        |

|                                                  |  |  |  |  |
|--------------------------------------------------|--|--|--|--|
| <b>Valeur de l'entreprise (actif économique)</b> |  |  |  |  |
|--------------------------------------------------|--|--|--|--|

# Example

Based on the following information, determine:

1/ The enterprise value (economic asset value)

|                                 | Entreprise A | Entreprise B | Entreprise C | Entreprise D |
|---------------------------------|--------------|--------------|--------------|--------------|
| <b>Capitalisation boursière</b> | 16 000       | 3 000        | 4 000        | 36 000       |
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| <b>REX (EBIT)</b>               | 960          | 96           | 300          | 1 600        |

|                                                  |        |       |       |        |
|--------------------------------------------------|--------|-------|-------|--------|
| <b>Valeur de l'entreprise (actif économique)</b> | 19 000 | 2 750 | 6 000 | 49 000 |
|--------------------------------------------------|--------|-------|-------|--------|

Valeur de l'entreprise = capitaux propres + dette financière nette

# Example

Based on the following information, determine:

2/The valuation multiples

- EV / Revenue
- EV / EBITDA
- EV / EBIT

|                          | Entreprise A | Entreprise B | Entreprise C | Entreprise D |
|--------------------------|--------------|--------------|--------------|--------------|
| Capitalisation boursière | 16 000       | 3 000        | 4 000        | 36 000       |
| Dette financière nette   | 3 000        | (250)        | 2 000        | 13 000       |
| Chiffre d'affaires       | 16 000       | 1 200        | 5 000        | 40 000       |
| EBE (EBITDA)             | 1 280        | 120          | 400          | 2 400        |
| REX (EBIT)               | 960          | 96           | 300          | 1 600        |

|                                           |        |       |       |        |
|-------------------------------------------|--------|-------|-------|--------|
| Valeur de l'entreprise (actif économique) | 19 000 | 2 750 | 6 000 | 49 000 |
|-------------------------------------------|--------|-------|-------|--------|

|        |  |  |  |  |
|--------|--|--|--|--|
| VE/CA  |  |  |  |  |
| VE/EBE |  |  |  |  |
| VE/REX |  |  |  |  |



# Example

Based on the following information, determine:

2/The valuation multiples

- EV / Revenue
- EV / EBITDA
- EV / EBIT

|                                 | Entreprise A | Entreprise B | Entreprise C | Entreprise D |
|---------------------------------|--------------|--------------|--------------|--------------|
| <b>Capitalisation boursière</b> | 16 000       | 3 000        | 4 000        | 36 000       |
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|                                                  |        |       |       |        |
|--------------------------------------------------|--------|-------|-------|--------|
| <b>Valeur de l'entreprise (actif économique)</b> | 19 000 | 2 750 | 6 000 | 49 000 |
|--------------------------------------------------|--------|-------|-------|--------|

|               |       |       |       |       |
|---------------|-------|-------|-------|-------|
| <b>VE/CA</b>  | 1,19  | 2,29  | 1,20  | 1,23  |
| <b>VE/EBE</b> | 14,84 | 22,92 | 15,00 | 20,42 |
| <b>VE/REX</b> | 19,79 | 28,65 | 20,00 | 30,63 |

# Valuing a company with multiples

Once the multiples are calculated for the companies in the sample, it is necessary to calculate the average and/or the median of the sample to obtain a multiple to apply to the company under study.

Sometimes large disparities may appear within the sample, making it useful to verify that the sample is relevant. It may also be appropriate to remove the maximum and minimum values from the sample.

Another option is to perform linear regressions to verify that the sample is relevant. If some companies in the sample appear to have irrelevant ratios, they should be excluded.

# Example

Based on the following information, determine:

- 1/ The enterprise value of the company under study according to the three multiples
- 2/ The equity value

|                                           | Entreprise A | Entreprise B | Entreprise C | Entreprise D | Entreprise étudiée |
|-------------------------------------------|--------------|--------------|--------------|--------------|--------------------|
| Capitalisation boursière                  | 16 000       | 3 000        | 4 000        | 36 000       | ?                  |
| Dette financière nette                    | 3 000        | (250)        | 2 000        | 13 000       | 800                |
| Chiffre d'affaires                        | 16 000       | 1 200        | 5 000        | 40 000       | 11 000             |
| EBE (EBITDA)                              | 1 280        | 120          | 400          | 2 400        | 880                |
| REX (EBIT)                                | 960          | 96           | 300          | 1 600        | 660                |
|                                           |              |              |              |              |                    |
| Valeur de l'entreprise (actif économique) | 19 000       | 2 750        | 6 000        | 49 000       | ?                  |
|                                           |              |              |              |              |                    |
| VE/CA                                     | 1,19         | 2,29         | 1,20         | 1,23         | ?                  |
| VE/EBE                                    | 14,84        | 22,92        | 15,00        | 20,42        | ?                  |
| VE/REX                                    | 19,79        | 28,65        | 20,00        | 30,63        | ?                  |

# Example

Based on the following information, determine:

1/ The enterprise value of the company under study according to the three multiples

|                          | Entreprise A | Entreprise B | Entreprise C | Entreprise D | Entreprise étudiée |
|--------------------------|--------------|--------------|--------------|--------------|--------------------|
| Capitalisation boursière | 16 000       | 3 000        | 4 000        | 36 000       | ?                  |
| Dette financière nette   | 3 000        | (250)        | 2 000        | 13 000       | 800                |
| Chiffre d'affaires       | 16 000       | 1 200        | 5 000        | 40 000       | 11 000             |
| EBE (EBITDA)             | 1 280        | 120          | 400          | 2 400        | 880                |
| REX (EBIT)               | 960          | 96           | 300          | 1 600        | 660                |

  

|                                           |        |       |       |        |   |
|-------------------------------------------|--------|-------|-------|--------|---|
| Valeur de l'entreprise (actif économique) | 19 000 | 2 750 | 6 000 | 49 000 | ? |
|-------------------------------------------|--------|-------|-------|--------|---|

  

|        |       |       |       |       |       |
|--------|-------|-------|-------|-------|-------|
| VE/CA  | 1,19  | 2,29  | 1,20  | 1,23  | 1,48  |
| VE/EBE | 14,84 | 22,92 | 15,00 | 20,42 | 18,29 |
| VE/REX | 19,79 | 28,65 | 20,00 | 30,63 | 24,77 |

# Example

Based on the following information, determine:

1/ The enterprise value of the company under study according to the three multiples

|                                                    | Entreprise A | Entreprise B | Entreprise C | Entreprise D | Entreprise étudiée |
|----------------------------------------------------|--------------|--------------|--------------|--------------|--------------------|
|                                                    |              |              |              |              | <i>Moyenne</i>     |
| VE/CA                                              | 1,19         | 2,29         | 1,20         | 1,23         | 1,48               |
| VE/EBE                                             | 14,84        | 22,92        | 15,00        | 20,42        | 18,29              |
| VE/REX                                             | 19,79        | 28,65        | 20,00        | 30,63        | 24,77              |
| Valeur de l'entreprise à partir du multiple VE/CA  |              |              |              |              | 16 236             |
| Valeur de l'entreprise à partir du multiple VE/EBE |              |              |              |              | 16 099             |
| Valeur de l'entreprise à partir du multiple VE/REX |              |              |              |              | 16 345             |

# Example

Based on the following information, determine:

2/ The equity value

|                          | Entreprise A | Entreprise B | Entreprise C | Entreprise D | Entreprise étudiée |
|--------------------------|--------------|--------------|--------------|--------------|--------------------|
| Capitalisation boursière | 16 000       | 3 000        | 4 000        | 36 000       | ?                  |
| Dette financière nette   | 3 000        | (250)        | 2 000        | 13 000       | 800                |
| Chiffre d'affaires       | 16 000       | 1 200        | 5 000        | 40 000       | 11 000             |
| EBE (EBITDA)             | 1 280        | 120          | 400          | 2 400        | 880                |
| REX (EBIT)               | 960          | 96           | 300          | 1 600        | 660                |

  

|                                           |        |       |       |        |         |
|-------------------------------------------|--------|-------|-------|--------|---------|
| Valeur de l'entreprise (actif économique) | 19 000 | 2 750 | 6 000 | 49 000 | ?       |
|                                           |        |       |       |        | Moyenne |
| VE/CA                                     | 1,19   | 2,29  | 1,20  | 1,23   | 1,48    |
| VE/EBE                                    | 14,84  | 22,92 | 15,00 | 20,42  | 18,29   |
| VE/REX                                    | 19,79  | 28,65 | 20,00 | 30,63  | 24,77   |

  

|                                                    |        |
|----------------------------------------------------|--------|
| Valeur de l'entreprise à partir du multiple VE/CA  | 16 236 |
| Valeur de l'entreprise à partir du multiple VE/EBE | 16 099 |
| Valeur de l'entreprise à partir du multiple VE/REX | 16 345 |

# Example

Based on the following information, determine:

2/ The equity value

|                          | Entreprise A | Entreprise B | Entreprise C | Entreprise D | Entreprise étudiée |
|--------------------------|--------------|--------------|--------------|--------------|--------------------|
| Capitalisation boursière | 16 000       | 3 000        | 4 000        | 36 000       | ?                  |
| Dette financière nette   | 3 000        | (250)        | 2 000        | 13 000       | 800                |
| Chiffre d'affaires       | 16 000       | 1 200        | 5 000        | 40 000       | 11 000             |
| EBE (EBITDA)             | 1 280        | 120          | 400          | 2 400        | 880                |
| REX (EBIT)               | 960          | 96           | 300          | 1 600        | 660                |

  

|                                           |        |       |       |        |         |
|-------------------------------------------|--------|-------|-------|--------|---------|
| Valeur de l'entreprise (actif économique) | 19 000 | 2 750 | 6 000 | 49 000 | ?       |
|                                           |        |       |       |        | Moyenne |
| VE/CA                                     | 1,19   | 2,29  | 1,20  | 1,23   | 1,48    |
| VE/EBE                                    | 14,84  | 22,92 | 15,00 | 20,42  | 18,29   |
| VE/REX                                    | 19,79  | 28,65 | 20,00 | 30,63  | 24,77   |

  

|                                                    |        |
|----------------------------------------------------|--------|
| Valeur de l'entreprise à partir du multiple VE/CA  | 16 236 |
| Valeur de l'entreprise à partir du multiple VE/EBE | 16 099 |
| Valeur de l'entreprise à partir du multiple VE/REX | 16 345 |

  

|                        |     |
|------------------------|-----|
| Dette financière nette | 800 |
|------------------------|-----|

  

|                                                        |        |
|--------------------------------------------------------|--------|
| Valeur des capitaux propres à partir du multiple VE/CA | 15 436 |
| Valeur de l'entreprise à partir du multiple VE/EBE     | 15 299 |
| Valeur de l'entreprise à partir du multiple VE/REX     | 15 545 |

# Example

|                          | Entreprise A | Entreprise B | Entreprise C | Entreprise D | Entreprise étudiée |
|--------------------------|--------------|--------------|--------------|--------------|--------------------|
| Capitalisation boursière | 16 000       | 3 000        | 4 000        | 36 000       | ?                  |
| Dette financière nette   | 3 000        | (250)        | 2 000        | 13 000       | 800                |
| Chiffre d'affaires       | 16 000       | 1 200        | 5 000        | 40 000       | 11 000             |
| EBE (EBITDA)             | 1 280        | 120          | 400          | 2 400        | 880                |
| REX (EBIT)               | 960          | 96           | 300          | 1 600        | 660                |

  

|                                           |        |       |       |        |         |
|-------------------------------------------|--------|-------|-------|--------|---------|
| Valeur de l'entreprise (actif économique) | 19 000 | 2 750 | 6 000 | 49 000 | ?       |
|                                           |        |       |       |        | Moyenne |
| VE/CA                                     | 1,19   | 2,29  | 1,20  | 1,23   | 1,48    |
| VE/EBE                                    | 14,84  | 22,92 | 15,00 | 20,42  | 18,29   |
| VE/REX                                    | 19,79  | 28,65 | 20,00 | 30,63  | 24,77   |

  

|                                                    |        |
|----------------------------------------------------|--------|
| Valeur de l'entreprise à partir du multiple VE/CA  | 16 236 |
| Valeur de l'entreprise à partir du multiple VE/EBE | 16 099 |
| Valeur de l'entreprise à partir du multiple VE/REX | 16 345 |

  

|                        |     |
|------------------------|-----|
| Dette financière nette | 800 |
|------------------------|-----|

  

|                                                        |        |
|--------------------------------------------------------|--------|
| Valeur des capitaux propres à partir du multiple VE/CA | 15 436 |
| Valeur de l'entreprise à partir du multiple VE/EBE     | 15 299 |
| Valeur de l'entreprise à partir du multiple VE/REX     | 15 545 |

Using trading multiples, we obtain a valuation range between 15,299 and 15,545.



# Transactions multiples

The logic is different with transaction multiples, transaction multiples correspond to the multiples observed during a company sale transaction :

- The sale price corresponds to the equity value :
- By adding net financial debt, we obtain the enterprise value
- Using the different financial aggregates, we obtain the valuation multiples

Transaction multiples are generally higher than trading multiples because they include a control premium.

However, this method may be limited by the quality of information available on truly comparable transactions.

## Asset-based (patrimonial) method

The asset-based method consists of valuing separately all assets, subsidiaries, and business segments of a company, and then deducting net financial debt to obtain the equity value.

This approach generally applies to conglomerates or groups with diversified activities.